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Astronomical forcing in Upper Miocene continental sequences: implications for the Geomagnetic Polarity Time Scale

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Abstract

We present an integrated stratigraphic study of cyclically bedded distal alluvial fan to lacustrine deposits in the late Miocene continental sections of Cascante and Cañizar (Teruel basin, NE Spain). The cyclostratigraphic analysis reveals that different scales of sedimentary cyclicity are present with a thickness ratio of about 1:2:5. Spectral analysis of colour records in the depth domain indicates the presence of a significant peak at ~2.2 m, which corresponds to the average thickness of the basic, mudstone–carbonate, cycle. Other peaks correspond to the large-scale cycle, which consists of clusters of five basic cycles, and to a cycle twice the average thickness of the basic cycle. Magnetostratigraphic results, in combination with small-mammal biostratigraphy, indicate that the three normal polarity intervals recorded in our sections correspond to C5n.2n, C5n.1n and C4Ar.2n. Assigning the ages of CK95 to the polarity reversals implies significant changes in sedimentation rate, which is not in agreement with the regularity of the sedimentary cyclicity. Thus, spectral analysis of high-resolution colour records in the time domain only produce a spectrum that is consistent with Milankovitch climate forcing, if several of the assigned age tie points are excluded. This indicates that the sedimentary cyclicity in these sections is related to astronomical variations in climate and that the ages of reversal boundaries are in error. Hence, we have calculated the astronomical durations for C4Ar.2n (87 ky), C4Ar.3r (54 ky), C5n.1n (141 ky), and C5n.1r (33 ky), which indeed show significant discrepancies with CK95. The duration pattern of our polarity intervals is confirmed by seafloor anomaly profiles and magnetostratigraphic records of deep-sea cores. Consequently, this study demonstrates that astronomically forcing in continental sequences can be a powerful tool to improve the fundamental dating of the geological record.
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Keywords: cyclostratigraphy; magnetostratigraphy, subchrons; spectral analysis; astronomical forcing; late Miocene

1. Introduction

The Astronomical Polarity Time Scale (APTS) for the late Neogene and Pleistocene is based on the tuning of cyclic marine sedimentary successions with a reliable magnetostratigraphy to astronomical target curves [1–4]. There are, however, several gaps in this time scale in particular because middle Miocene marine successions lack a reliable magnetostratigraphyic record. This problem can be solved by using cyclic continental successions such as the Orera Composite Section (OCS) in northeastern Spain, which has an excellent magnetostratigraphic record

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[5,6]. The astronomical origin of sedimentary cycles in the OCS was convincingly demonstrated and astronomical ages were obtained for all middle to late Miocene polarity reversals between C5Ar and C5n. These astronomical ages revealed significant discrepancies with the ages (and duration) of the subchrons in the Geomagnetic Polarity Time Scale (GPTS) of Cande and Kent [7] (CK95), but appeared to be in very good agreement with ages derived from seafloor spreading rates [5].

Another important gap in the marine APTS is present between C5n and C4r. For this interval, the magnetostratigraphy is admittedly weak in the Mediterranean [8] and completely lacking at Ceara Rise [2]. Although astronomical ages have been presented, these ages are either based on a poor magnetostratigraphy (Monte Gibliscemi, Sicily [8]), on poorly defined reversal boundaries (Monte dei Corvi [9]), or have been estimated by exporting astronomically tuned bioevents to extra-Mediterranean records having a reliable magnetostratigraphy (Monte Gibliscemi and ODP Site 845 [10]). In this paper, we present the integrated stratigraphic results of the continental Cascante and Cañizar sections, which encompass the interval between the youngest part of Chron C5n.2n and C4Ar. We will first show that these sections, of which the strata are nearly horizontal, provide a reliable magnetostratigraphy, which can be calibrated to the GPTS using small-mammal biostratigraphic information. Second, we will demonstrate the astronomical origin of the sedimentary cyclicity and attempt to establish its phase relation with the orbital parameters. Consequently, the duration of subchrons can be established by counting the number of cycles per polarity interval. Finally, the new floating chronology for this part of the GPTS will finally be discussed through a detailed comparison with seafloor anomaly profiles and magnetostratigraphic results from marine DSDP/ODP sites that cover the same stratigraphic interval.

2. Geological setting

The Teruel basin is a 100-km-long and 15-km-wide NNE–SSW trending semigraben, dissecting older NW–SE Paleogene compressional structures of the Iberian Chain [11]. The basin has a complicated tectonic structure related to a NNE–SSW fault lineation that developed during the Neogene, the reacti-

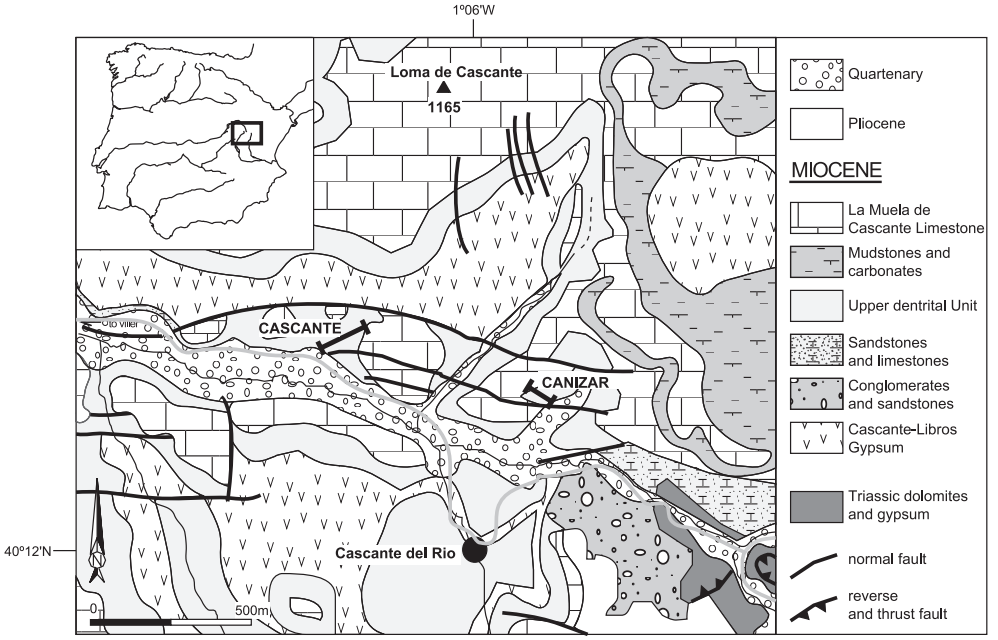


Fig. 1. Geologic and location map of the Cascante and Cañizar sections (modified after Ref. [14]).

vation of the pre-Neogene NW–SE structural trend of the Iberian Chain, and the diapiric activity of Triassic evaporites [12,13]. The sedimentary sequence, deposited in an internal drainage basin, is dominated by alluvial red beds, lacustrine/palustrine carbonates and gypsum. The coarse-grained red bed successions at the basin margin pass laterally into fine-grained distal alluvial fan and mudflat deposits and to shallow lacustrine carbonate and gypsum deposits in the axial zone of the graben [13].

The study area is located south of Teruel, along the Villel-Cascante del Rio road, and is flanked by Mesozoic rocks, predominantly Triassic siliciclastics and evaporites and Jurassic carbonates (Fig. 1). The sections Cascante and Cañizar (Fig. 1) are characterised by a distinct sedimentary cyclicity of alternating red to green siliciclastic mudstones and carbonates. These cycles were first reported by Bakx [15]. They can be ascribed to the Upper detrital unit of the Cascante-Libros area [16]. The top of the sections is capped by a thick carbonate sequence of rhythmically bedded shallow lacustrine/palustrine carbonates with thin intercalations of greenish-gray marls. They correspond to the Muela de Cascante Limestones of Anadón et al. [16].

3. Cyclostratigraphy

3.1. Cascante section

The Cascante section shows a distinct cyclic bedding of red and/or green mottled siliciclastic mudstones alternating with white carbonates (Fig. 2). The siliciclastic mudstone beds have an average thickness of 1.8 m (range 1.4–3.3 m) and show a variety of colours: dark red at the base, gradually changing to orange colours with gray-green mottling, and finally to green colours with yellow mottling at the top. Siliciclastic mudstone beds are massive and often contain gypsum aggregates/nodules and occasionally clastic or carbonate pebbles. The carbonate beds have an average thickness of 0.5 m (range 0.2–0.9 m). Centimetre-thick organic-rich layers are present at the base of most carbonate beds and contain plant debris, fossil remains of gastropods, and (rare) small mammal bones. The carbonate beds show a wide variety of characteristics; they are (poorly) laminated or massive, strongly consolidated or weakly consolidated, the bed tops may be bioturbated by rootlets, and they contain variable amounts of fossil remains of gastropods, ostracods and charophyte stems. In several carbonate beds, micronodular gypsum can be found, while gypsum may also be present in fillings of animal burrows or in laminae showing an alternation with micritic carbonate. Generally, gypsum becomes more abundant in the top parts of carbonate beds.

The very distinct mudstone–carbonate cycle is regarded as the basic small-scale cyclicity in the section (Fig. 2). Carbonate beds mark the base of a sedimentary cycle and are numbered in ascending order. Possible extra cycles are denoted with the suffix ‘A’ following the number of the preceding cycle. The average cycle thickness in Cascante is 2.2 m, but the cycles are slightly thicker in the lower part than in the upper part of the section. In total, 33 small-scale cycles and two ‘A’ cycles have been recognised. A second type of cyclicity, the large-scale cycle, is defined by clusters of four to five small-scale cycles delimited by thick carbonate beds (Fig. 2). Such clusters are clearly discernable between the thick carbonate beds of cycles 9, 14 and 18 and, although less obvious, between the thick carbonate beds of cycles 5 and 9, and 18 and 23.

In the upper part of the Cascante section, a transition from regularly developed mudstone–carbonate cycles to a thick limestone unit is observed. This transition is gradual, although the base of this unit is slightly undulating. The onset to the transition is marked by a conspicuously red interval, which corresponds to cycles 28 to 30. Cycle 30 is the last cycle in Cascante that displays a distinct purple to red colour with a sharp colour change to green siliciclastic mudstone. The subsequent cycles 31–33 are characterised by an alternation of 0.7-m-thick limestone beds with 1-m-thick marls. The upper limestone unit (only the base is shown in Fig. 2) is thick, monotonous, but cyclically bedded, and is best exposed in a steep cliff about 200 m east of the section. A conspicuously thick tabulated bed is present at the base of the unit, which splits into three carbonate units (separated by a red siliciclastic interval) to the east of Cascante, in the direction of the paleomargin of the basin.

3.2. Cañizar section

The tabulated limestone, at the base of the upper unit, is also distinguished in the Cañizar area, where it

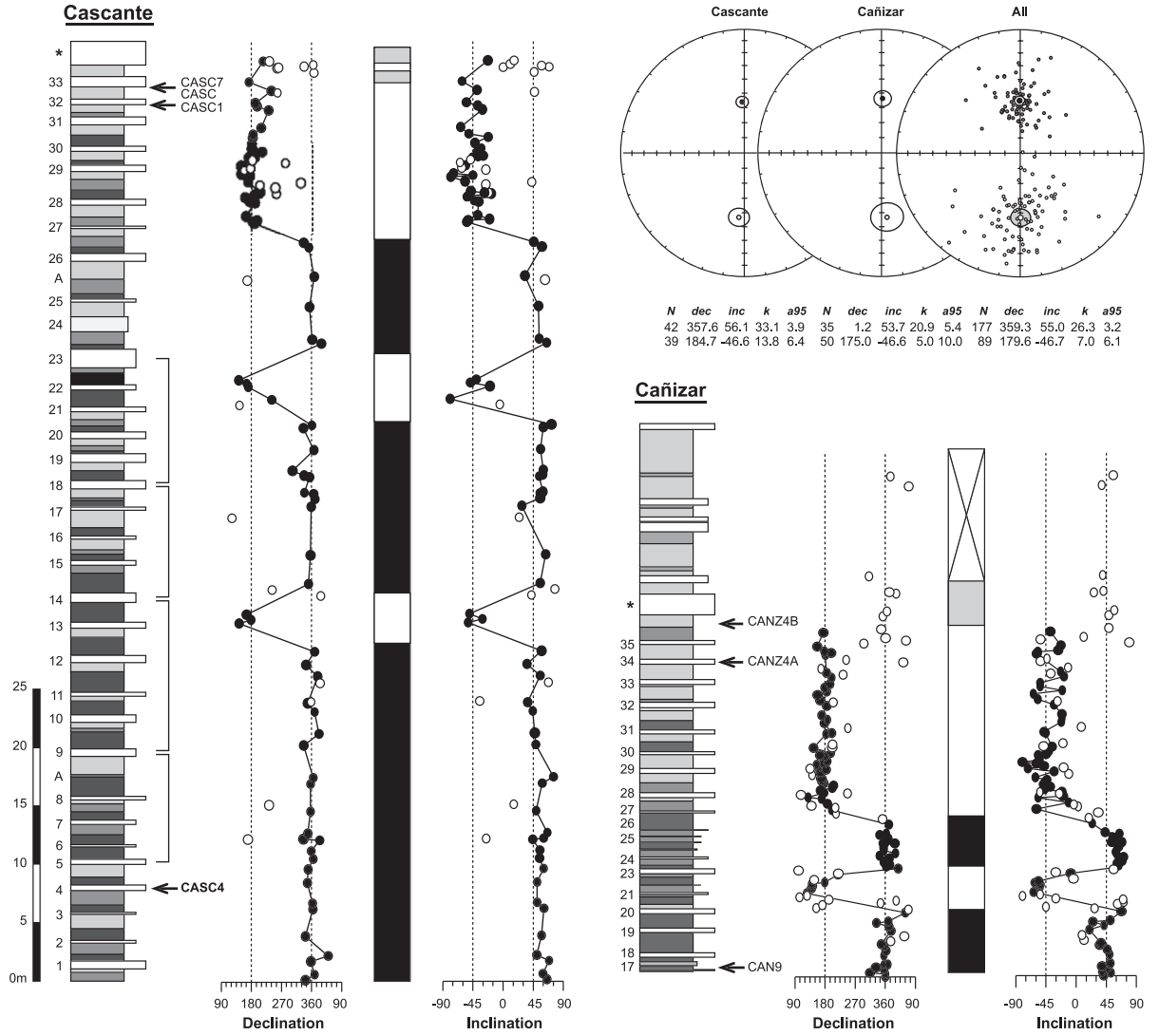


Fig. 2. Lithology columns, paleomagnetic results and polarity columns of the Cascante and Cañizar sections. In the lithology columns, light/medium/dark gray shades correspond to green(ochre)/orange mottled/red siliciclastic mudstones and white to carbonate beds, respectively. Protruding carbonate beds illustrate the degree of consolidation; the more protruded, the more consolidated the beds are. The asterisk corresponds to the tabulated limestone (see text for details). The location of mammal sites is shown with arrows; the cycle numbers are indicated to the left of the lithologic column; the brackets show the large-scale cycles (see text). In the polarity columns, black (white) zones indicate normal (reversed) polarity. Gray-shaded zone represents uncertain and the cross noninterpreted polarity. The open (closed) symbols denote reliable (unreliable) ChRM directions. In the upper part, equal area projections of the reliable ChRM directions for the Cascante, Cañizar sections and their combined results are shown. The 95% confidence ellipse for the normal and reverse directions is indicated. No bedding correction is applied (strata are practically horizontal). Statistical information includes number of samples (*N*), declination (Dec.), inclination (Inc.), Fisher's parameter (*k*), radius of the 95% confidence cone ($\alpha 95$).

separates two distinct units (Fig. 2); a well-exposed cyclic lower unit of about 30 m thick and a poorly exposed upper unit of about 15 m thick that lacks distinct cyclicity. The cyclic lower unit can be subdivided into an upper and lower part. The upper part comprises 10 mudstone–carbonate cycles. The silici-

clastic mudstone beds have an average thickness of 1.4 m (0.8–1.7 m) and are mostly ochre and green coloured and only rarely red. The carbonate beds have an average thickness of 0.35 m (0.25–0.5 m), are massive, sometimes laminated, and are rich in fossil remains of mainly gastropods. Thin organic-rich layers of fossil gastropods and plant debris are often preserved below and within these beds. The cyclicity in the lower part is less distinct. Nevertheless, at least 10 cycles of alternating red silt to sandstone and caliche beds can be distinguished (Fig. 2). The average thickness of the red silt to sandstone beds is similar to that of the siliciclastic mudstone beds in the upper part of the section. Subtle colour variations occur ranging from dark red at the base to more (orange) red with yellow and (bluish) gray mottles at the top of the beds. The caliche beds have an average thickness of 0.25 m (0.03–0.4 m) and often show red to orange colours with yellow mottling.

A direct, bed-to-bed, correlation between the Cascante and Cañizar sections was hampered by the presence of faults and complicated by lateral facies changes, especially in the thick carbonate unit. Nevertheless, a correlation was accomplished by matching cycle patterns, which could later be tested by the magnetostratigraphic results. The cycle pattern in the upper part of Cañizar corresponds fairly well with the patterns in Cascante; the conspicuous purple to red cycle 30 confirms this correlation. Several discrepancies in the cycle patterns of both sections are, however, observed as well. Especially in the lower part of Cañizar where red colours prevail and the cyclicity is characterised by alternating red silt/sandstones and caliches. Moreover, two additional cycles are present in Cañizar, just below the tabulated limestone bed. The thick tabulated limestone bed in Cascante is, however, laterally divided into three limestone beds intercalated with marl, which may indicate that here the cycles are less distinct, caused by a more distal position in the basin. Alternatively, it could also be interpreted that erosion has removed two cycles in Cascante.

3.3. Quantifying the cyclicity

To test the astronomical origin of sedimentary cycles in the Cascante section, it is necessary to determine the regularity and continuity of the sedimentary cyclicity, and to determine the dominant periods—in terms of cycles per m—that underlie this cyclicity. Therefore, we applied spectral analysis on colour records of the Cascante section in the depth domain. Colour measurements were performed on fresh surface exposures and were taken at an average of 16 levels per cycle using a portable photospectrometer (Minolta CM508i). The measurements are automatically converted into the reflectance values $L^*a^*b^*$. The $L^*a^*b^*$ (or CIELAB) value is a uniform colour scale recommended by the Commission Internationale de l'Eclairge (CIE). L^* defines lightness, a^* denotes the red/green value and b^* the yellow/blue value. In our study, we used the L^* and a^* reflectance records because they best display the cyclicity as observed in the field. Low L^* values correspond to siliciclastic mudstones while high L^* values correspond to carbonate beds. As for the a^* record, low values correspond with carbonates and green coloured siliciclastic mudstones, high values with red siliciclastic mudstones (Fig. 3). For spectral analysis, the Blackman–Tukey (BT) and the Maximum Entropy (ME) methods were applied using the AnalySeries program [17]. The Gaussian bandpass filter used to extract the dominant spectral components was chosen such that it covers the maximum possible area of a peak in the power spectrum.

The power spectra of the L^* and a^* records show dominant peaks between 2.2 and 2.7 m (Fig. 3). The range of these values approximately corresponds to the average thickness of the basic mudstone–carbonate cycles. The extracted 2.2- and 2.7-m components follow the basic small-scale cyclicity, whereby the 2.2-m component mainly follows the slightly thinner cycles in the upper part and the 2.7-m component the thicker cycles in the lower part. As both components correspond to the basic cycles, we have chosen to centre the bandpass filter at 2.38 and 2.39 m with a bandwidth that covers both the 2.2- and 2.7-m components (Fig. 3). Another significant peak recorded in the power spectra of the a^* and, although less significant, of the L^* record is around 4.2 m (Fig. 3). The bandpass filters set at 4.2 m for both records show that the extracted components follow (increased/reduced) amplitude variations, which correspond to thickness differences of successive carbonate and red siliciclastic mudstone beds (Fig. 3). This is clearly seen in cycle interval 6–11 and 23–31.

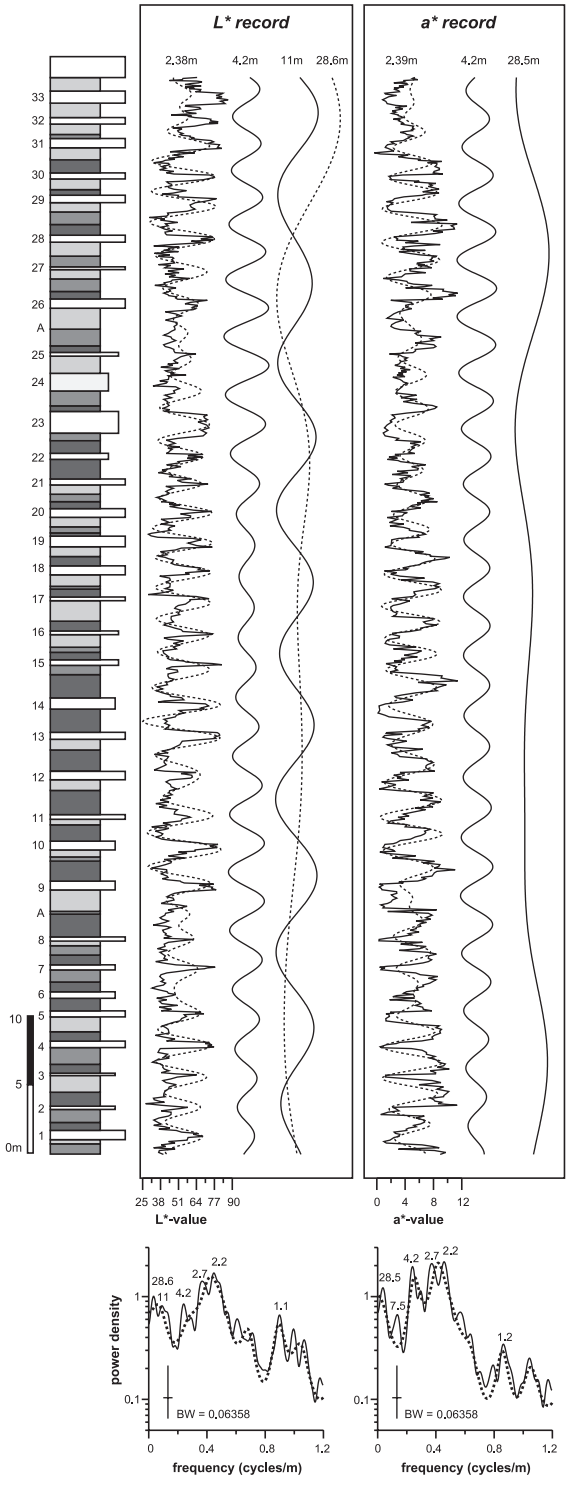


Fig. 3. Colour records (L^* and a^*) of the Cascante section and results of spectral analysis of the colour records in the depth domain. Also shown are the filtered components that correspond to the spectral peaks. In the power spectra, the Blackman–Tukey (solid) and Maximum Entropy (dashed) results are shown. The numbers at the peaks denote the cycle thickness in m. The band widths (BW) and upper and lower confidence intervals (80%) are shown in the lower left-hand corner. The bandpass widths of the filtered components for L^* are centred at 0.42 ± 0.095 (2.38 m), 0.238474 ± 0.03 (4.2 m), 0.09221 ± 0.0225 (11 m) and at 0.034974 ± 0.0222 (28.6 m); and for a^* at 0.417 ± 0.089 (2.38 m), 0.238474 ± 0.0265 (4.2 m) and at 0.034976 ± 0.025 (29 m). All results were obtained from the AnalySeries program of Paillard et al. [17]. Note that the 1.1 and 1.2 m cycles are an artefact and represent the harmonic of the 2.2-m-thick basic mudstone–carbonate cycle.

In the low-frequency domain, spectral peaks occur at 11 and 28.6 m for the L^* and at 28.5 m for the a^* power spectrum (Fig. 3). The 11-m extracted component follows the prominent thick carbonate beds which define the clusters of cycles of the large-scale cyclicity (Fig. 3). Finally, a 28.5-m component is distinguished in the a^* record power spectrum. The extracted component shows a more or less similar, but opposite, relationship with lithology as the 28.6-m component in the L^* record and marks the conspicuous red siliciclastic mudstone transition zone and the carbonate-green marl cycles in the top part of Cascante, respectively (Fig. 3).

The results demonstrate that the sedimentary cyclicity in Cascante is regular and continuous. The ratio between these different scales of cycles is 1:2:5, which suggests an astronomically forced cyclicity by the 21-ky precession, the 41-ky obliquity and the 100-ky eccentricity cycle [18,19]. An accurate age model must be established, however, to prove that this astronomical interpretation of the sedimentary cyclicity is indeed correct. For this purpose, we used magnetostratigraphy in combination with small-mammal biostratigraphy.

4. Mammal biostratigraphy

The Cascante and Cañizar sections produced a total of seven small-mammal faunules (Fig. 2). The levels are not very productive, and the preservation is poor. Detailed taxonomic descriptions will be given in a future paper.

The lowermost locality Cascante 4 is situated at the base of the carbonate bed of cycle 4 and yielded 10 species. This locality can be assigned to Local Zone I [20] based on the presence of the hamster *Megacricetodon debruijni* and the dormouse *Ramys multicrestatus*, which are exclusively confined to this Zone, and the absence of true mice (Murinae) whose regular occurrence defines the base of the next younger zone J [20–22]. Zone I roughly correlates to European Mammal Neogene unit 9 (MN9), defined by the fauna of Can Llobateres in the Vallès-Penedès Basin [23].

The level of Cañizar 9 in cycle 17 yielded three teeth, including a damaged murine molar which we have identified as *Progonomys* cf. *hispanicus*. A preliminary assignment to Zone J1 can be made following the definition of Zone J, which base is defined by the First Regular Occurrence of *P. hispanicus* [22]. More fossil material is needed to establish if this species is indeed regular at this location, thereby verifying whether or not the assignment to Zone J1 is correct.

The other small mammal fossils are found higher up in the sections in cycles 31 and 32, the latter cycle represented by two laterally equivalent sites (Cascante, Cascante 7). In Cañizar, two sites are located in cycles 34 and 35, Cañizar 4A and 4B respectively. The presence of *P. hispanicus* (Murinae) in all these sites is diagnostic for Local Zone J. The primitive state of the molars practically excludes the possibility of subzone J3 [24]. An allocation to either subzone J1 or J2 is confirmed by the presence of the shrews *Crusafontina endemica* and *Miosorex grivensis*, which are not known from the area in levels higher than J2. Subzone J2 is slightly more probable because of the presence of the mole form *Desmanella* in Cascante 1 (cycle 32). This form is not recorded in any of the other J1 localities in the area [22]. The levels in cycles 31–35 correlate with MN10 which is defined by the reference fauna of Masía del Barbo 2B, a J2 fauna in the Teruel area itself [13,22,23].

5. Magnetostratigraphy

5.1. Methods

Two standard paleomagnetic cores were sampled per mudstone–carbonate cycle in the Cascante section. Most of the cores were taken from the red siliciclastic mudstones because this lithology is most suitable for paleomagnetic analysis. Better exposure and a less weathered surface in the upper part of Cascante allowed more detailed sampling. In the Cañizar section, at least five cores per cycle were taken at an average distance of 35 cm. The poorly exposed top unit above the Cañizar section proved to be unsuitable for drilling and only a few cores were obtained. At least one core specimen per sampling level was thermally demagnetised using stepwise temperature increments of 30–50 °C. The natural remanent magnetisation (NRM) was measured on a horizontal 2G Enterprises DC SQUID cryogenic magnetometer (noise level 3×10^{-12} A m^2). NRM directions were determined using principal component analysis.

5.2. Paleomagnetic results

The thermal demagnetisation diagrams from both Cascante and Cañizar are generally of good quality, although some carbonates gave less reliable results. The NRM intensity of the specimens from Cascante ranges between 100 and 10,000 μA/m, but is slightly lower (15–1000 μA/m) at Cañizar. Zijderveld diagrams generally comprise three components (Fig. 4). The first is a randomly oriented component, which is removed at temperatures around 100 °C. The second component of normal polarity is removed at temperatures of 250–280 °C and most probably represents a viscous overprint inherited from the present-day field. The third component of dual polarity is removed at higher temperatures and is interpreted as the characteristic remanent magnetisation (ChRM). For the red siliciclastic mudstones, the ChRM is usually removed at temperatures above 600 °C to a maximum of 690 °C, indicating unblocking temperatures for hematite (Fig. 5a and b). By contrast, the ChRM of almost half of the red (silty) siliciclastic mudstones of Cañizar is already removed at temperatures between 520 and 570 °C, which is a typical unblocking temperature for magnetite. The ChRM of green (and yellow-mottled) siliciclastic mudstones and of carbonates is removed between 400 and 600 °C, which probably indicates the presence of magnetite as well (Fig. 5c and d). The initial NRM intensities of the latter lithologies are much lower than in the red siliciclastic mudstones,

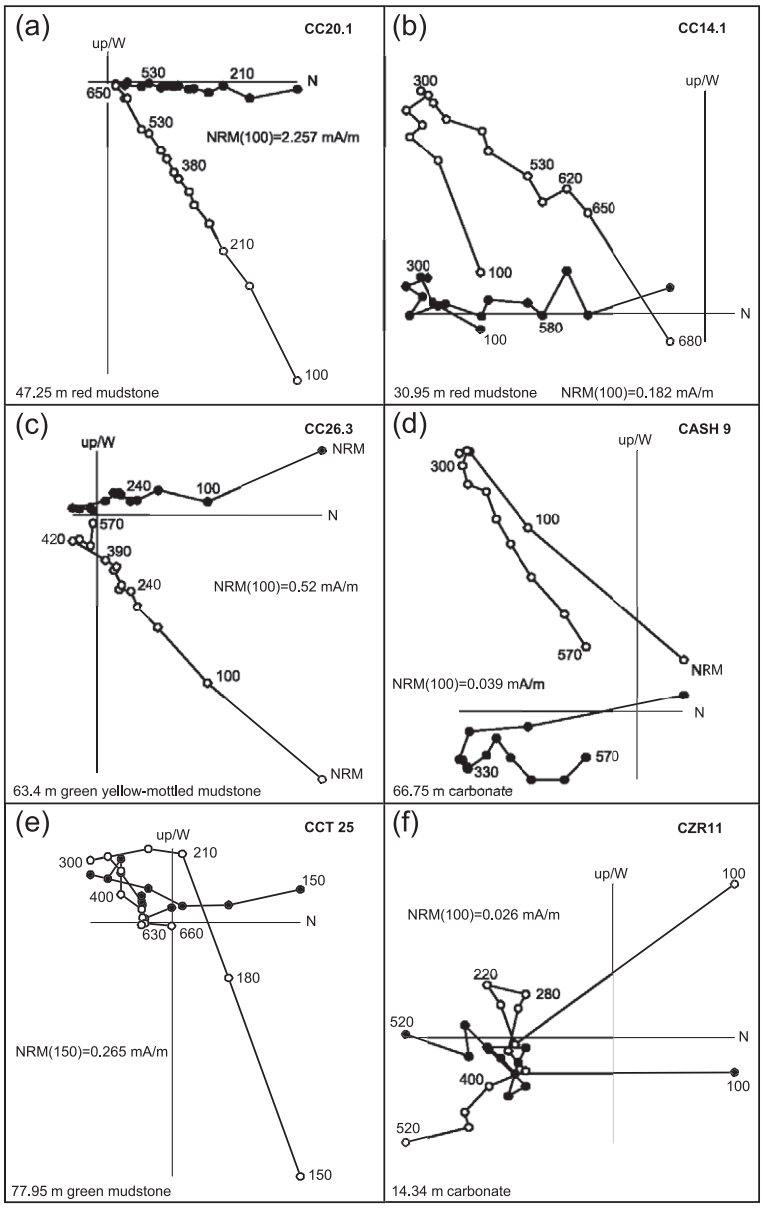


Fig. 4. (a–f) Zijderveld diagrams of thermally demagnetised specimens from the Cascante (CC, CASH, and CCT) and Cañizar (CZR) sections. Closed (open) symbols represent the projection of vector endpoints on the horizontal (vertical) plane; values denote temperature steps in °C; stratigraphic level and type of lithology are indicated in the lower left-hand corner. See text for details.

probably due to lower concentrations of magnetic minerals. Nevertheless, at least half of all the measured specimens from the green siliciclastic mudstones and the carbonates show good quality diagrams (Fig. 4d and e). The other half of the carbonates, including the thick monotonous unit at the top of the sections, produced noninterpretable and very weak polarity directions (Fig. 4c and f).

The ChRM directions and polarity zones show that six polarity intervals are recorded in Cascante and

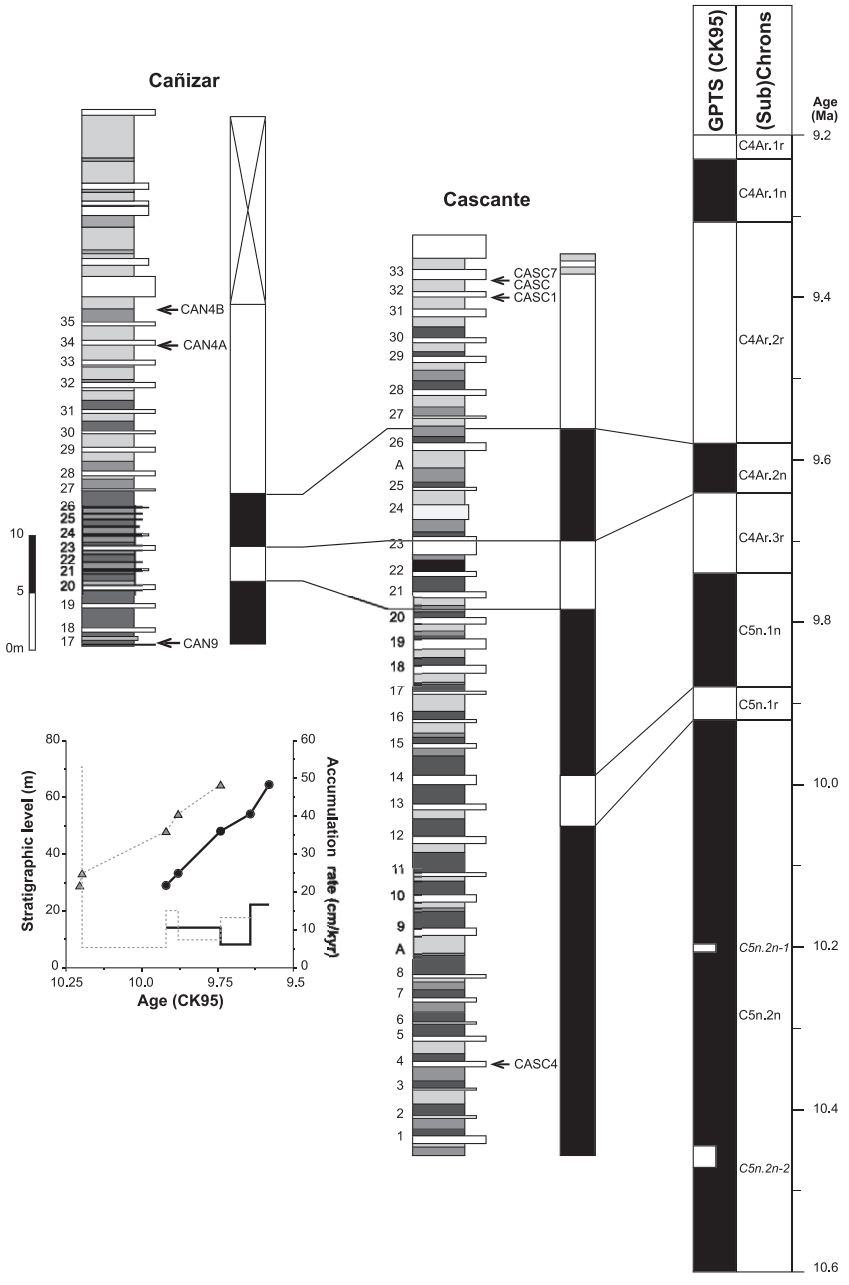


Fig. 5. Calibration of the Cascante and Cañizar magnetostratigraphy to the geomagnetic polarity time scale (GPTS) [7]. The diagram shows the accumulation rates for the Cascante section and for the alternative age model (dashed lines) as discussed in footnote 1 (see also Fig. 2 caption).

four in Cañizar (Fig. 2). The paleomagnetic results from the uppermost limestone unit is poor, due to unclear polarity signals. In Cascante, one specimen in the carbonate and one in the green siliciclastic mudstone of cycle 33 show respectively normal and reversed directions (Fig. 5e). In Cañizar, however, there is no indication for a normal polarity in cycle 33. Some diagrams of samples from the tabulated lime-

stone bed also seem to indicate normal directions, but they do not show a gradual decay towards the origin. Hence, we consider these normal directions in the upper limestone unit unreliable.

6. Correlation to the GPTS

The magnetostratigraphic results clearly confirm the cyclostratigraphic correlation between the two sections (Fig. 5). The number of sedimentary cycles within polarity intervals in Cañizar equals that of Cascante, and the polarity reversals are located at exactly the same cyclostratigraphic position. The dominantly normal polarities, combined with the small-mammal data indicating MN 9 and MN 10, strongly suggest a correlation to the upper part of Chron C5n of the GPTS [25]. The characteristic polarity pattern indicates that the three normal polarity intervals most likely correlate to C5n.2n, C5n.1n and C4Ar.2n, respectively (Fig. 5). This correlation allows us to calculate the sedimentation rate per polarity interval, using the ages of CK95 for the polarity reversals. These calculations indicate significant changes in sedimentation rate throughout the section (Fig. 5 inset), which is certainly not in agreement with the very regular sedimentary cyclicity. Nevertheless, other magnetostratigraphic correlations result in even greater changes, suggesting that these options are less likely. Comparing the polarity patterns of Cascante and CK95 (Fig. 5), the observed changes in sedimentation rate are obviously caused by differences in the relative length of the reversed and normal intervals that correlate to C4Ar.3r and C4Ar.2n. The most logical explanation is that the position of the polarity reversal C4Ar.2n

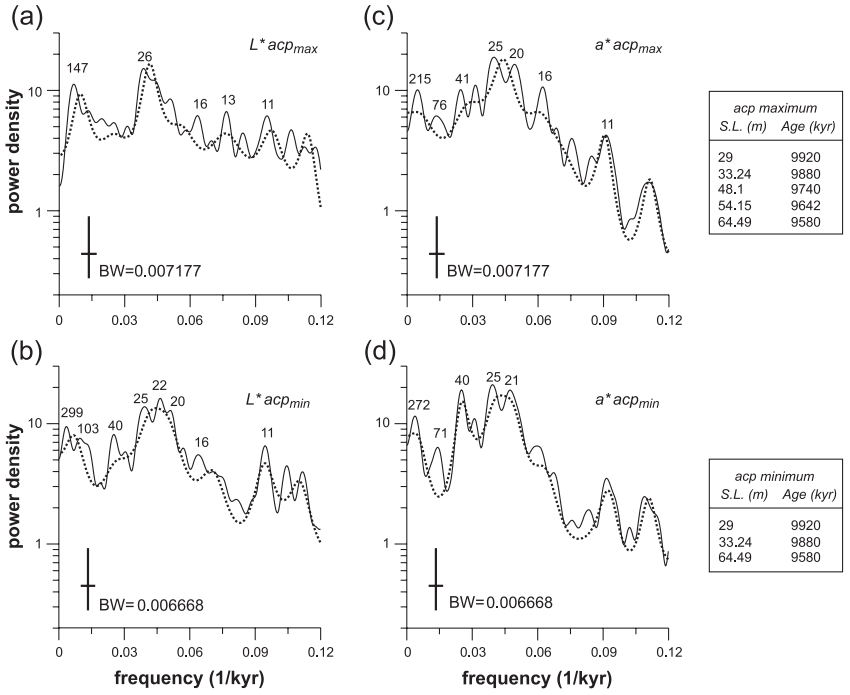


Fig. 6. (a–d) Spectral analysis results of colour time series (L^* and a^*) of the Cascante section according to the maximum and the minimum age calibration points (see text for details). The numbers at the spectral peaks denote the period in ky. In the lower left corners, the bandwidths and the upper and lower confidence (80%) intervals are shown. The power density spectra were computed with the AnalySeries program using the Blackman–Tukey (solid line) and Maximum Entropy (dashed line) methods [17]. Tables show the stratigraphic levels (SL) and corresponding age calibration points (in ky) used for constructing time series and subsequent spectral analyses. The ages are according to the Geomagnetic Polarity Time Scale of Cande and Kent [7].

(o) has not been correctly determined in our sections, but the reliable paleomagnetic signal and the perfect agreement between Cascante and Cañizar makes this option unsatisfactory to us. The alternative explanation is that the pattern in CK95, which is derived from marine magnetic anomaly profiles of the ocean floor, is not correct. These anomaly profiles are in some cases difficult to interpret, especially when a series of very short polarity intervals are located on relatively slow spreading plates. In that case, less problems will be expected with the position of the upper and lower reversals of these intervals, which can be regarded as more reliable.

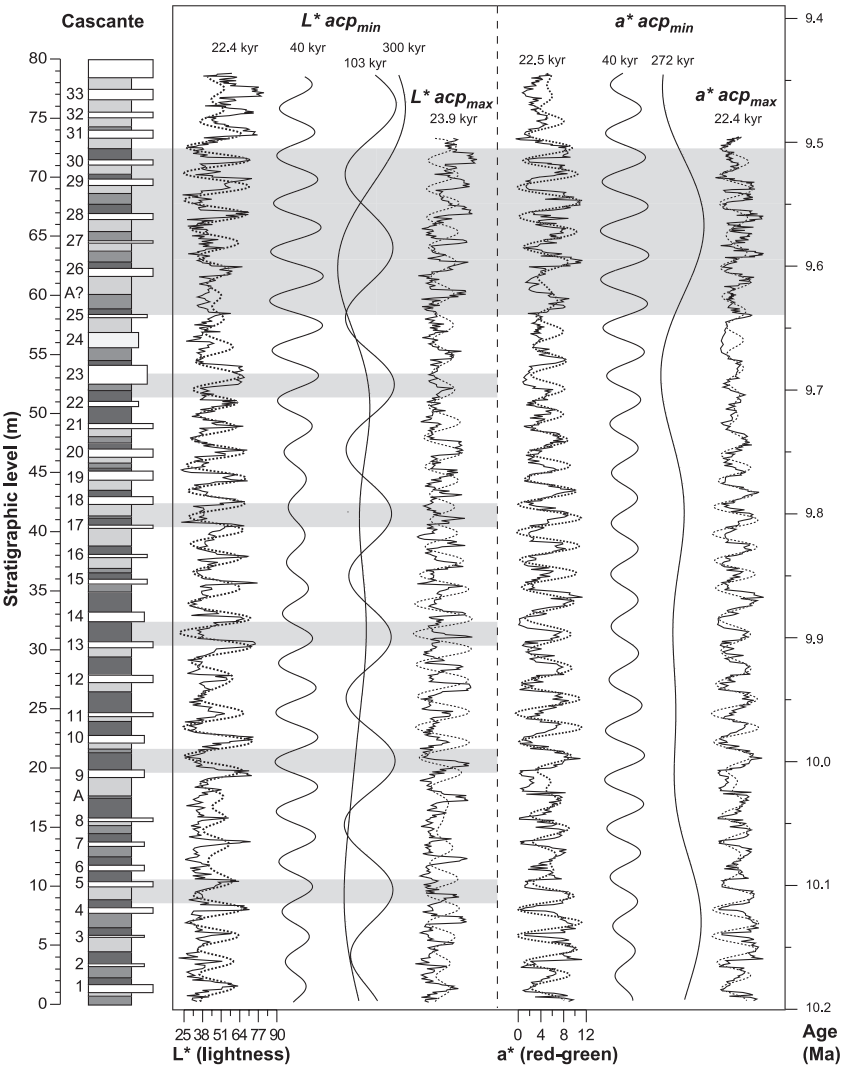


Fig. 7. Filtered records of the L^* and a^* colour time series according to acp_{min} and acp_{max} with a minimum and a maximum number of age tie points, respectively (see text for details). The narrow gray shaded zones define the limits of the large-scale cyclicity while the broad shaded zone indicates the conspicuous red siliciclastic mudstone transition interval. The bandpass width of the filtered components for L^*acp_{max} is centred at 0.04186 ± 0.0062 (23.9 ky); for L^*acp_{min} at 0.0446 ± 0.0095 (22.4 ky), 0.025042 ± 0.0025 (40 ky), 0.009683 ± 0.002 (103 ky), and 0.003339 ± 0.002 (300 ky); for a^*acp_{max} at 0.044722 ± 0.0962 (22.4 ky) and for a^*acp_{min} at 0.044407 ± 0.00932 (22.5 ky), 0.025543 ± 0.003175 (39 ky) and at 0.03673 ± 0.00266 (272 ky).

6.1. Spectral analysis in the time domain

The calibration of the Cascante magnetostratigraphy to CK95 provided age control points for constructing time series of the colour records by means of linear interpolation between, and extrapolation beyond, the tie points. Two time series were generated for both L^* and a^* colour records using a minimum (acp_{min}) and a maximum (acp_{max}) number of calibration points, by excluding and including ages for C4Ar.2n (o) and C5n.1n (y), respectively (see Fig. 6).[1]

Using the minimum number of calibration points, the L^*acp_{min} spectrum shows dominant peaks between 20 and 25 ky while other peaks can be distinguished at around 40, 103 and 300 ky (Fig. 6a). The bandpass filters for the L^*acp_{min} record show that the 22.4-ky component follows the small-scale cyclicity suggesting an orbital control by precession (Fig. 7). The 40-ky component follows alternating thick–thin carbonate beds dominantly present in the upper part of Cascante (between cycles 25 and 33). This variation in bed thickness is most likely related to the obliquity cycle (Fig. 7). Finally, the 103-ky component follows the large-scale cyclicity suggesting a link with the 100-ky eccentricity cycle (Fig. 7).

The power spectrum of a^*acp_{min} also reveals significant periods between 21 and 25 ky, around 40 and 272 ky (Fig. 6b). The extracted 22.5-ky component follows the small-scale cycles while the 40-ky filtered component follows alternating thick–thin beds of red siliciclastic mudstone mainly between cycles 23 and 31 (Fig. 7). This suggests a precession and obliquity origin for these cycles, respectively. The 272-ky filtered component follows the dark (dominantly red) intervals in the lower and upper part of the Cascante section (Fig. 7).

[1] An alternative correlation to the GPTS was earlier discussed by Abdul Aziz [26] and involved a correlation of the upper reversed polarity interval to subchron C4Ar.3r and the short reversed polarity between cycles 13 and 14, to 'cryptochron' C5n.2n-1. The sedimentation rate profiles of this option, however, showed less consistent results (Fig. 5). Spectral analysis of the time series according to this alternative calibration revealed a significant period at around 42 ky, suggesting that the basic small-scale cyclicity is controlled by the obliquity cycle. This seems highly unlikely, especially because the 1:2:5 ratio was lacking and no coherency was found with the results from the depth domain.

Spectral analysis results of the L^*acp_{max} time series yields a spectrum with dominant peaks at around 26 and 147 ky, while the most significant periodicity in the a^*acp_{max} spectrum is found between 20 and 25 ky (Fig. 6a). We centred the Guassian bandpass filter at 23.9 ky, to obtain a maximum coverage of the peak at 26 ky in the L^*acp_{max} spectrum. As expected, the filtered component does not follow the basic small-scale cyclicity (Fig. 7). The bandpass filter for the a^*acp_{max} record, centred at 22.4 ky, only follows the small-scale cycles in the lower part of Cascante (cycles 1–23).

The spectral analysis results thus clearly demonstrate that the factor controlling the outcome of the analyses in the time domain is related to the age calibration points used for constructing the time series. The best fit with astronomical forcing comes from the acp_{min} colour spectra, which produce a 1:2:5 ratio. This is in perfect agreement with the results obtained in the depth domain. More importantly, however, the acp_{min} spectral results reveal the presence of $\sim$20-, 40- and 100-ky periodicities, which unambiguously demonstrates the astronomical origin of the sedimentary cycles in Cascante. In contrast, the acp_{max} spectra lack this ratio. This indicates that we will seriously have to consider inaccuracies in the magnetic anomaly profiles of the C5n to C4r interval of the GPTS.

7. Discussion

7.1. Astronomical origin of the cycles and phase relations

The sedimentary cyclicity in the Cascante and Cañizar sections is comparable with the cyclicity observed in the Orera Composite Section (OCS) of the Calatayud basin [6]. In the OCS, the different scales of cycles were also demonstrated to correspond to the astronomical parameters [5]. Proving the astronomical origin of the sedimentary cyclicity in Cascante is important as well, because the cycles allow a more comprehensible interpretation of lake-level variations and climate change than the sedimentary cycles described from the slightly older OCS section, which were deposited in an extraordinary environment [27].

The siliciclastic mudstones and silt–sandstones in the study area are interpreted as alluvial deposits [28,29] that accumulated during lake-level lowstands

on the distal areas of the alluvial fan. The carbonate beds represent lacustrine and palustrine deposits that accumulated in a very shallow lake during lake-level highstands, which developed after submergence of the distal plains of the alluvial fan. The final stage of carbonate precipitation is marked by the presence of gypsum in the top parts of carbonate beds indicating a change towards more saline lake conditions. Thus, the cyclicity in the studied sections is the result of an alternation between distal alluvial fan and shallow lake environments, which likely was controlled by waxing and waning of the lake. Alonso-Zarza and Calvo [30] established a general paleoclimatic reconstruction of similar and time equivalent deposits from the northern part of the Teruel basin. They suggested, based on lithological characteristics and on the environmental interpretation of the carbonates, that carbonate deposition occurred during periods of increased precipitation.

In the marine realm, periods of increased precipitation are associated with sapropels [31,32] and correspond to precession minima and, consequently, to summer insolation maxima [33]. In contrast, homogeneous marls are deposited during periods of reduced seasonality and more arid climate conditions at times of precession maxima. Extrapolating these Mediterranean phase relations to the continental mudstone–carbonate cycles of Cascante and assuming that lake-level highstands correspond to wetter climate conditions, would imply that carbonates are formed during more humid climate conditions at times of precession minima. The mudstones on the other hand are deposited at times of decreased precipitation. These conditions are reached during precession maxima. The effect of eccentricity is such that it modulates the precession amplitude. This implies that the thick carbonate beds in the Cascante section, which define the clusters of five small-scale cycles, correspond to (prolonged) periods of maximum amplitude variations in precession (minima) that occur during 100-ky eccentricity maxima. Less distinct carbonate beds will correspond to periods of reduced amplitude variations in precession (minima) during 100-ky eccentricity minima.

Ideally, the sedimentary cycles in the Cascante section should be tuned to the astronomical target curves as has been done for the cycles in the Orera Composite Section (OCS) [5]. However, the phase relation with eccentricity is complicated by the great variety in internal build-up of the carbonate beds in the Cascante and Cañizar sections. This is perhaps best illustrated by our first attempts to tune the cycles to the astronomical target curves. Starting form the magnetostratigraphic age constraints, the best fit with the astronomical curves was obtained when the thick carbonate beds correspond to eccentricity minima, instead of eccentricity maxima. Therefore, we first have to better understand the origin and nature of the different types of carbonates and, subsequently, their phase relation with eccentricity. In addition, the lateral consistency of the cycle patterns, especially the pattern related to obliquity and eccentricity, needs to be confirmed in other sections of the area before we can reliably use them for astronomical tuning. At this point, one may argue whether astronomically dated reversal boundaries in other sections, such as Monte Gibliscemi (Sicily)/ODP Site 845 [10] and Monte dei Corvi (Italy) [9], may serve as a reliable starting point for the astronomical tuning of the sedimentary cycles in Cascante. However, as stated before, the astronomical ages of the subchron boundaries in question are subject to considerable uncertainties, which strongly limit their use for a reliable tuning. Inevitably, these uncertainties will results in a tuning that critically depends on ill-founded assumptions about, in particular, the phase relation with eccentricity. For these reasons, we refrain from presenting a detailed tuning of the sedimentary cycles of Cascante to the astronomical target curves at this stage.

7.2. Duration of subchrons

Now that the precessional origin of the small-scale sedimentary cycles in the Cascante section has been proven, we can determine the duration of the polarity intervals by counting the number of precession-controlled cycles in each subchron and multiply them with the average period of precession [34] (Table 1). Comparison with CK95 shows that our astronomical duration of subchron C4Ar.3r is about two times shorter (54 vs. 98 ky), while subchron C4Ar.2n is somewhat longer (87 vs. 62 ky; Table 1). In addition, it appears that the astronomical duration of C4Ar.3r is about three times shorter than C5n.1n (54 vs. 141 ky), while this difference is much less in CK95 (98 vs. 140 ky). The duration of these subchrons in CK95 is based on the magnetic anomaly profiles of Blakely [35]. Blakely's profile for

Table 1
Astronomical duration of subchrons in Cascante compared with the duration according to the GPTS of CK95

Subchron	Number of cycles	Duration (ky)	
		Cascante	CK95
C4Ar.2n	3 1/2–4 1/2	87 ± 10	62
C4Ar.3n	2 1/2	54	98
C5n.1n	6 1/2	141	140
C5n.1r	1 1/2	33	40

The astronomical duration was calculated by using the average period of 21.7 ky per precession cycle [34].

the interval between C5 and C4A, however, is subject for revision because it is based on a few measured profiles with a short linear segment. This results in a poor control on the anomalies and therefore corrections must be applied for reconstructing the magnetic anomaly profile (D.S. Wilson, personal communication, 2001). Magnetic anomaly profiles from the Rivera-Pacific plate are composed of numerous profiles with good control on the linear segment, while the 'intermediate' motion of the Rivera-Pacific plate (half-rate of 35 km/My) points to a resolution of about 160 ky [36]. Nevertheless, the synthetic magnetic profile shows that the length of C5n.1n is indeed about 1.5–2 times longer than C4Ar.3r. A similar length ratio can also be assessed from high-resolution near-bottom magnetic data of the North and South Pacific [37], and from the magnetostratigraphic records of Deep Sea Drilling Program site 608 [38,39] and Ocean Drilling Program sites 1092 and 845 [40,41] (Fig. 8). In addition, the paleomagnetic results from the Monte dei Corvi section in northern Italy seem to confirm the ratio despite the fact that successive polarity intervals are marked by one or two samples only [9]. All these records support our magnetostratigraphic calibration and confirm our cyclostratigraphic observation that the sedimentation rate in Cascante and Cañizar was fairly constant

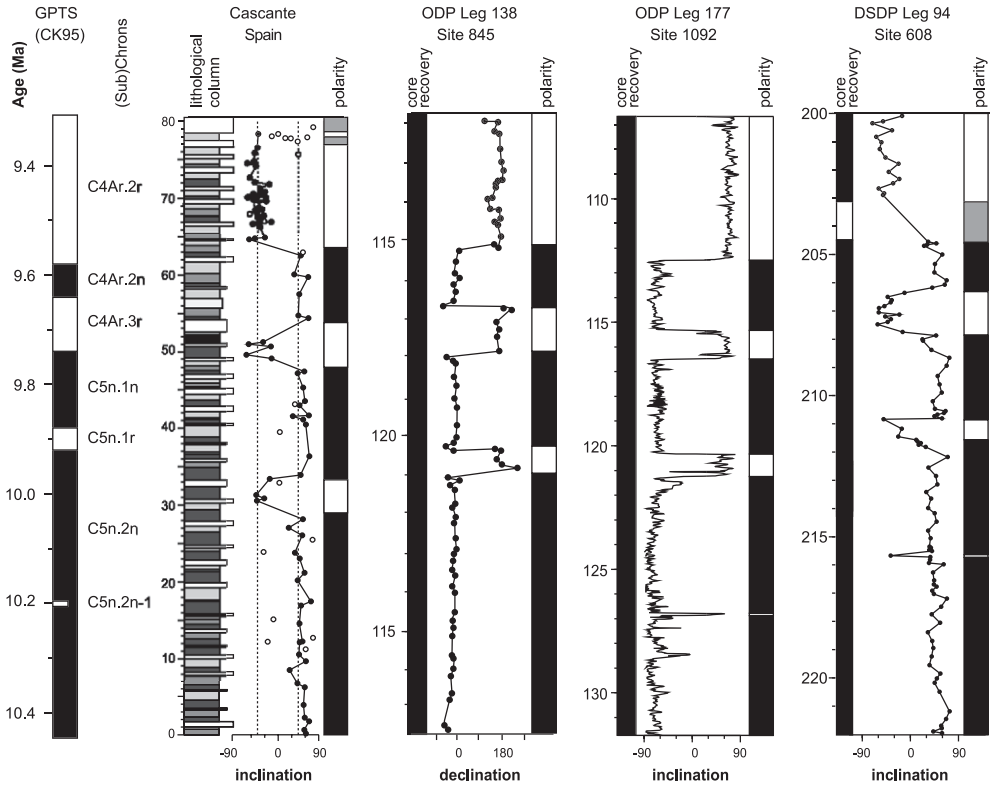


Fig. 8. Comparison between duration of subchrons in the Geomagnetic Polarity Time Scale (GPTS) and the length of polarity zones in the magnetostratigraphic records of Cascante, ODP/DSDP sites 845 [41], 1092 [40] and 608 [39].

throughout the entire cyclic part of the section. Thus, it appears that significant errors must have been introduced in the chronologies and sedimentation rate calculations of all previous DSDP and ODP records of the corresponding time interval (e.g., in Refs. [38,40,41]). Hence, we conclude that the late Miocene part of the GPTS will certainly have to be revised in the near future, either by a thorough restudy of the marine magnetic anomaly patterns or by high-resolution paleomagnetic sampling of marine (and continental) sedimentary sequences together with an accurate tuning of these sequences to the astronomical target curves.

8. Conclusions

The correlation of the Cascante and Cañizar magnetostratigraphy to CK95 is straightforward, but implies a significant change in sedimentation rate that is not acceptable from a cyclostratigraphic point of view. Time series analysis of colour records confirms the presence of superimposed sedimentary cycles that are related to the Earth's orbital parameters of precession, obliquity and eccentricity. An unambiguous tuning of the Cascante cycles to the astronomical target curves is hampered by uncertainties in the phase relation between thick carbonate beds and the 100-ky eccentricity cycle, in combination with the lack of reliable and precise ages for polarity reversals. Nevertheless, the number of precession controlled sedimentary cycles per polarity interval allowed us to determine the astronomical duration of the recorded subchrons. Our results indicate significant discrepancies in duration, especially for C4Ar.3r (54 vs. 98 ky) and C4Ar.2n (87 vs. 62 ky) as compared with CK95. Our interpretation is consistent with seafloor anomaly profiles and magnetostratigraphic records of deep-sea cores, indicating that the late Miocene GPTS should be thoroughly revised, especially for the interval between C5n and C4r.

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